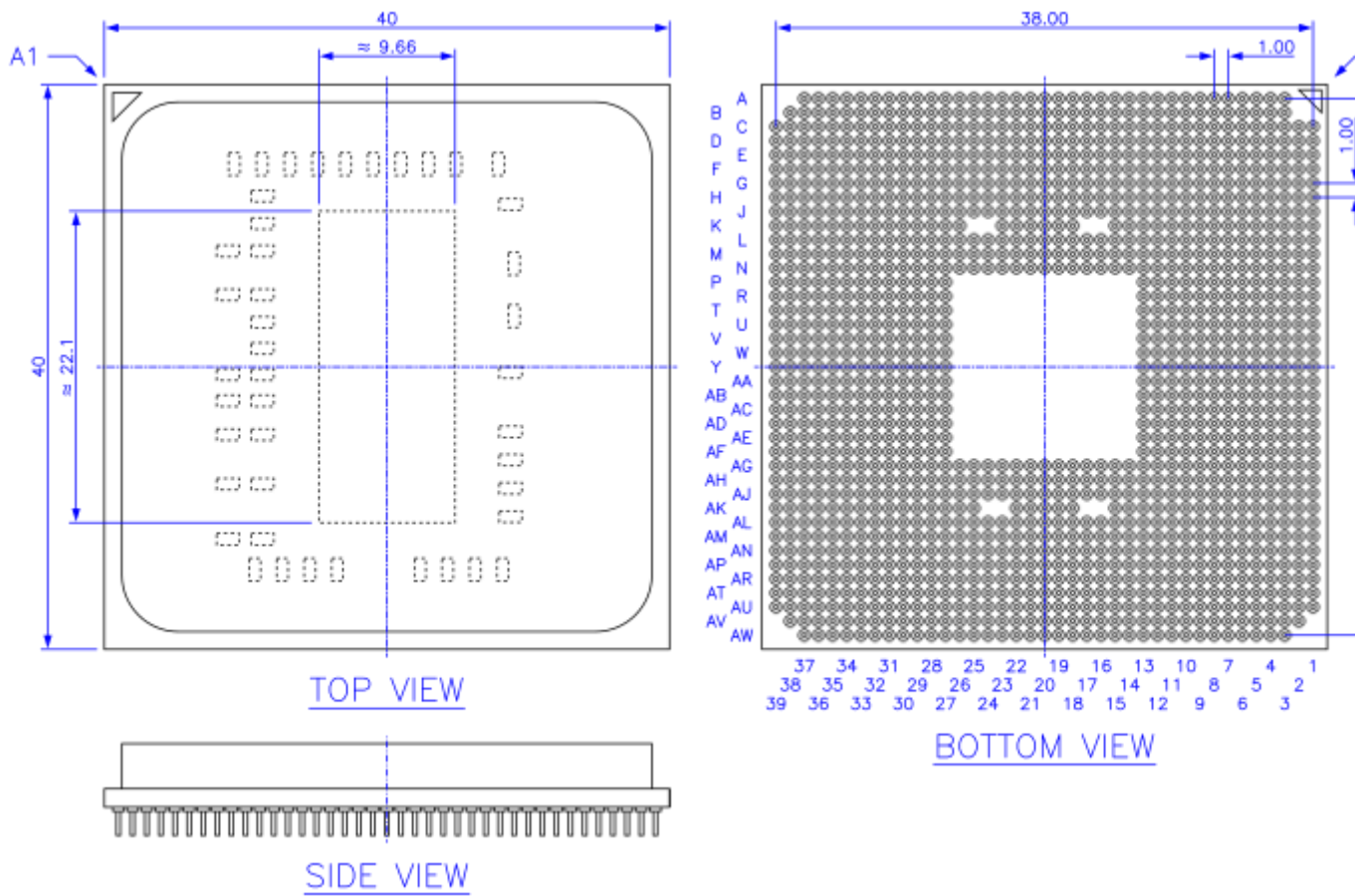
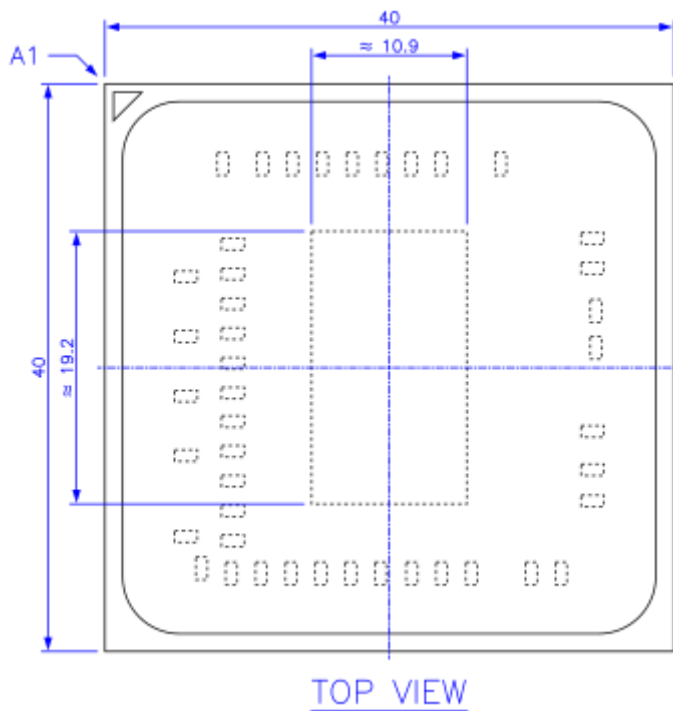


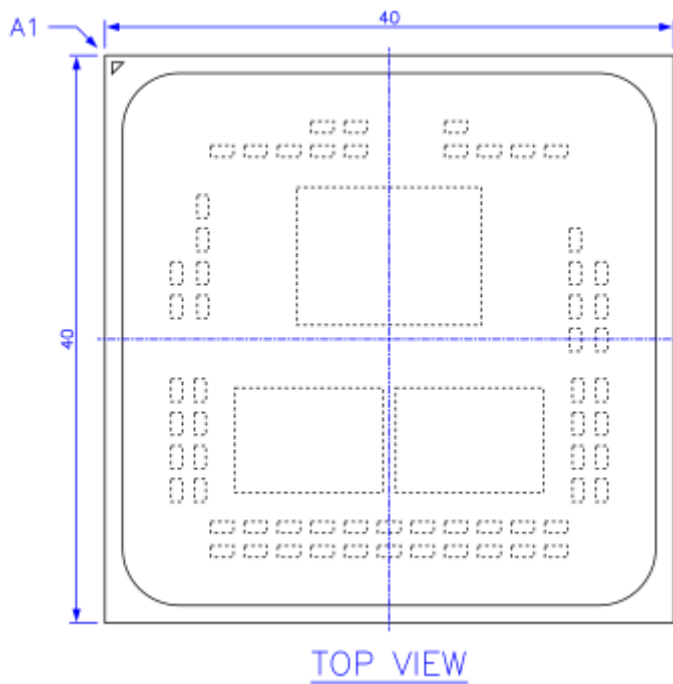
Package Diagrams[edit]



Approximate dimensions of the "Summit Ridge" ($\leq 8C$, Zen, 14 nm, Feb. 2017) and "Pinnacle Ridge" ($\leq 8C$, Zen+, 12 nm, April 2018) package. All dimensions in millimeters.

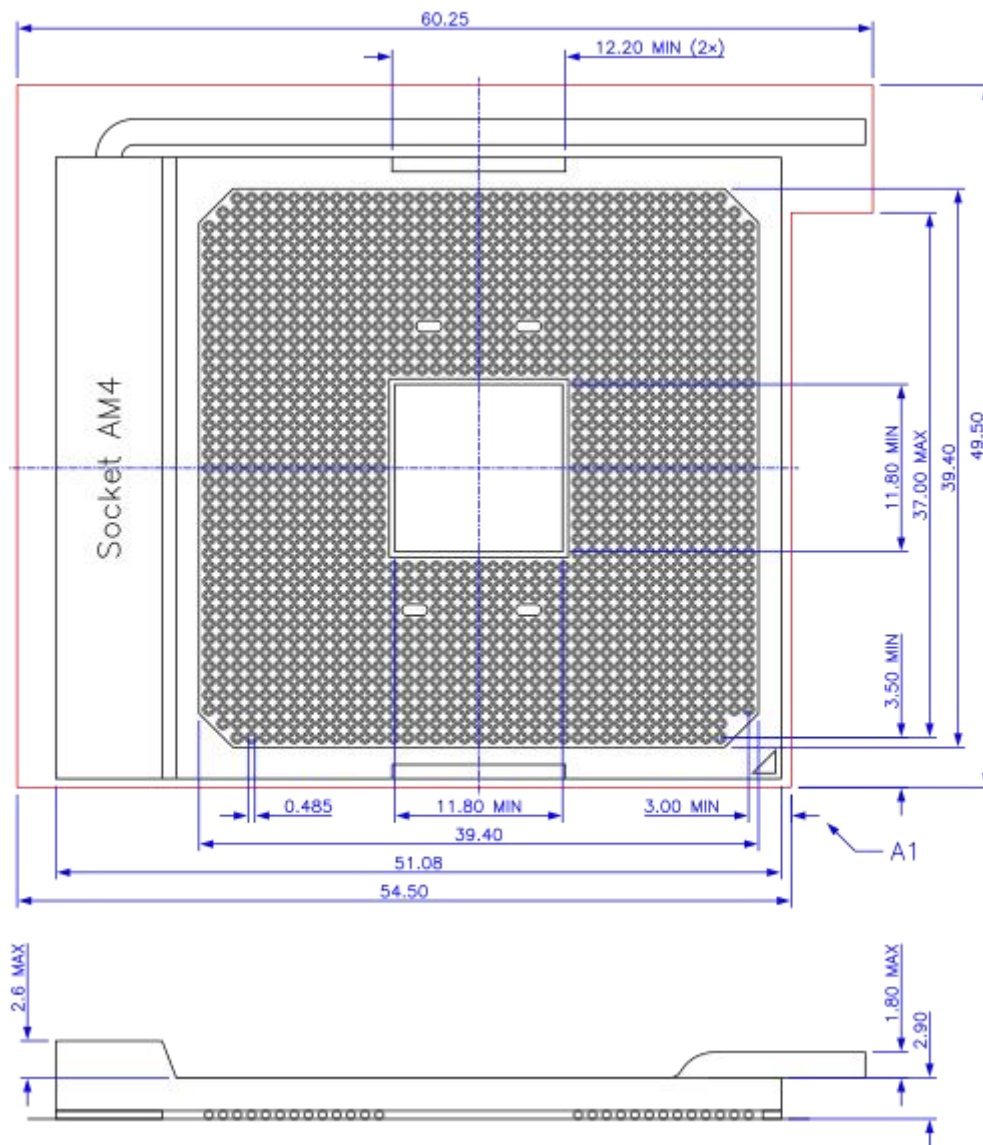


Approximate dimensions of the "Raven Ridge" desktop package ($\leq 4C/11CU$, Zen, 14 nm, Oct. 2017). Mobile and embedded versions are available in a FP5 package. All dimensions in millimeters.



Approximate dimensions of the "Matisse" package ($\leq 16C$, Zen 2, 7/14 nm, July 2019). The left CCD is unpopulated in $\leq 8C$ models. All dimensions in millimeters.

Socket Diagrams



No data available. The diagram shows the dimensions of Foxconn Interconnect Technology Part No. PZ1331A-51ZZ*-1H. All dimensions in millimeters.

Pin Description

Signal	Description
MA/MB_ACT_L	DRAM Channel A/B Activation Command
MA/MB_ADD[13:0]	DRAM Column/Row Address
MA/MB_ADD_17	DRAM Column/Row Address 17
MA/MB_ALERT_L	DRAM Alert (CRC error and Command/Address parity error)
MA/MB_BANK[1:0]	DRAM Bank Address
MA/MB_BG[1:0]	DRAM Bank Group
MA/MB_CAS_L_ADD[15]	DRAM Column Address Strobe or Column/Row Address 15

MA/MB_CHECK[7:0]	DRAM ECC Check Bits
MA/MB_DATA[63:0]	DRAM Data Bus
MA/MB_DM[8:0]	DRAM Data Mask
MA/MB_DQS_H/L[8:0]	DRAM Differential Data Strobe
MA/MB_EVENT_L	DRAM Thermal Event Status
MA/MB_PAROUT	DRAM Command/Address Parity
MA/MB_RAS_L_ADD[16]	DRAM Row Address Strobe or Column/Row Address 16
MA/MB_RESET_L	DRAM Reset
MA/MB_WE_L_ADD[14]	DRAM Write Enable or Column/Row Address 14
MA/MB_ZVDDIO_MEM_S3	
MA/MB_ZVSS	DRAM Interface Drive-Strength Auto-Compensation Resistor to VSS
MA0/MA1/MB0/MB1_CKE[1:0]	DRAM Channel A/B DIMM 0/1 Clock Enable
MA0/MA1/MB0/MB1_CLK_H/L[3:0]	DRAM Channel A/B DIMM 0/1 Differential Clock
MA0/MA1/MB0/MB1_CS_L[1:0]	DRAM Channel A/B DIMM 0/1 Chip Select
MA0/MA1/MB0/MB1_ODT[1:0]	DRAM Channel A/B DIMM 0/1 Enable Pin for On Die Termination
P_GFX_RXP/RXN[15:0]	PCIe GFX Receive Data Differential Pairs
P_GFX_TXP/TXN[15:0]	PCIe GFX Transmit Data Differential Pairs
P_GPP_RXP/RXN[3:0]	PCIe GPP Receive Data Differential Pairs
P_GPP_TXP/RXN[3:0]	PCIe GPP Transmit Data Differential Pairs
P_HUB_RXP/RXN[3:0]	PCIe Hub (chipset) Receive Data Differential Pairs
P_HUB_TXP/TXN[3:0]	PCIe Hub Transmit Data Differential Pairs
PCIE_RST_L	Reset signal for PCIe devices
P_ZVDDP	
P_ZVSS	
POA/POB_ZVSS	PCIe Drive-Strength Auto-Compensation Resistor to VSS

SATA_RX(0-1)P/N	SATA Receive Data Differential Pairs (alt. func. of P_GPP[2], P_GPP[3])
SATA_TX(0-1)P/N	SATA Transmit Data Differential Pairs
SATA_ACT_L	SATA Channel Active (HD LED)
SGPIO0_CLK	SGPIO Interface CLK Output; SGPIO supported on Socket AM4?
SGPIO0_DATAIN	SGPIO DATA Input
SGPIO0_DATAOUT	SGPIO DATA Output
SGPIO0_LOAD	SGPIO LOAD Output
SATA_IS(0-1)_L	
SATA_ZP(0-1)_L	Zero Power SATA ODD
SATA_ZVDDP	
SATA_ZVSS	
DP0-DP2_TXP/TXN[3:0]	DisplayPort 0-2 Main Link Differential Transmitter Lane 0-3
DP0-DP2_AUXP/AUXN	DisplayPort 0-2 Auxiliary Channel
DP0-DP2_HPD	DisplayPort 0-2 Hot Plug Detect Input
DP_BLON	Display Panel Backlight Enable
DP_DIGON	Display Panel Power Enable
DP_VARY_BL	Display Backlight Brightness Control
DP_STEREOSYNC	StereoSync output for shutter glasses
DP_AUX_ZVSS	
DP_ZVSS	
USB_HSD(0-3)P/N	USB Port 0-3 High Speed I/O Differential Pairs
USB_SS_(0-3)RXP/RXN	USB Port 0-3 Super Speed Receive Differential Pairs
USB_SS_(0-3)TXP/TXN	USB Port 0-3 Super Speed Transmit Differential Pairs
USB_OC(0-3)_L	USB Port 0-3 Over Current signal from USB connector
USB_SS_ZVDDP	

USB_SS_ZVSS	
USB_ZVSS	
USB(0-3)_ZVSS	USB Port 0-3 Drive-Strength Auto-Compensation Resistor to VSS
AZ_BITCLK	Azalia HD Audio Interface Bit Clock
AZ_RST_L	HDA Reset
AZ_SDIN(0-2)	HDA Serial Data Input from Codec 0-2
AZ_SDOUT	HDA Serial Data Output to Codec
AZ_SYNC	HDA Sync Signal to Codec
SPKR	PC speaker/beeper PWM output
SPI_CLK	SPI Clock
SPI_DO	SPI Data Out or Data 0 for multi-I/O SPI/eSPI device
SPI_DI	SPI Data In or Data 1
SPI_WP_L	SPI Write Protect or Data 2
SPI_HOLD_L	SPI Hold Signal (asserted low to hold the SPI transaction) or Data 3
SPI_TPM_CS_L	SPI Chip Select for TPM
SPI_CS1/CS2_L	SPI Chip Select
ESPI_CLK	ESPI Clock (alt. func. of SPI_CLK)
ESPI_DAT(0-3)	ESPI Data[0], Data[1:0], Data[3:0] (alt. func. of SPI_DO/DI/WP_L/HOLD_L)
ESPI_CS_L	ESPI Chip Select (SPI_CS2_L)
ESPI_ALERT_L	ESPI Alert Input (LDRQ0_L)
ESPI_RESET_L	ESPI Reset (KBRST_L)
LAD(0-3)	LPC Command/Address/Data
LDRQ0_L	Encoded DMA/Bus Master Request 0
LFRAME_L	LPC Bus Frame
LPCCLK(0-1)	LPC 33 MHz Clock

LPC_CLKRUN_L	LPC CLKRUN Signal
LPC_PD_L	LPC Power Down
LPC_PME_L	LPC Power Management Event
LPC_RST_L	LPC Reset
SERIRQ	Serial IRQ for DMA
I2C(2-3)_SCL	I ² C Port 2-3 Clock
I2C(2-3)_SDA	I ² C Port 2-3 Data
SCL0	SMBus Port 0 Clock (alt. func. of I2C2)
SDA0	SMBus Port 0 Data
SCL1	SMBus Port 1 Clock (alt. func. of I2C3)
SDA1	SMBus Port 1 Data
AGPIO*	Advanced GPIO pin for interrupt, wake, or I/O
EGPIO*	Enhanced GPIO for I/O only
GENINT(1-2)_L	Generic Interrupt Request
GFX_CLKP/N	PCIe GFX 100 MHz Differential Reference Clock
GPP_CLK0(0-3)P/N	PCIe GPP 100 MHz Differential Reference Clock
CLK_REQG_L	PCIe GFX Clock Request
CLK_REQ(0-3)_L	PCIe GPP Clock Request
OSCIN	14 MHz Clock Input
X32K_X1/X2	32768 Hz Clock XTAL for the integrated RTC
X48M_X1/X2	48 MHz Clock XTAL for the integrated clock generator
RTCCLK	32768 Hz Real Time Clock output, for a device requiring an RTC clock
X48M_OSC	48 MHz clock output for devices requiring a single-ended OSC input
KBRST_L	Keyboard Controller Reset Input (warm reset)
PWR_BTN_L	Power Button; Requests sleep state or causes wake event

PWR_GOOD	Power Good Input; Asserted when all voltages and clock inputs are within specification
PWROK	Power OK; Asserted by the processor after all power planes are active, the system clock generators are powered up and run stably
RESET_L	Bidirectional signal that resets the processor when asserted; Normally controlled by an internal state machine but can also be asserted by a second external source
RSMRST_L	Resume Reset from motherboard; Asserted on power up, deasserted when S5 power supplies are within specification
SOA3_GPIO	SOA3 Indicator
SLP_S3/S5_L	S3/S5 Sleep State Power Plane Control Signal
SYS_RESET_L	System Reset Input (reset button)
WAKE_L	PCIe WAKE_L signal, wake system out of sleep state
BLINK	Blink LED S-state Indicator
FANINO	Fan tachometer input
FANOUT0	Fan PWM output
ALERT_L	Programmable pin that can indicate different events, including a SB-TSI interrupt
PROCHOT_L	Asserted to force the processor into HTC -active state
SIC	Sideband Interface (SB-TSI) Clock
SID	Sideband Interface Data
THERMTRIP_L	Temperature Trip Input/Output
DBREQ_L	Debug Request input to JTAG controller
DBRDY	
TCK	JTAG Clock
TDI	JTAG Data Input
TDO	JTAG Data Output
TMS	JTAG Mode Select

TRST_L	JTAG Reset
TEST*	Test Pins
SVC	Serial VID Interface Clock
SVD	Serial VID Interface Data
SVT	Serial VID Interface Telemetry
VDDIO_AUDIO	Azalia HD Audio power supply
VDDBT_RTC_G	Integrated Real Time Clock battery power supply
VDDCR_CPU	Core power supply
VDDCR_CPU_SENSE	VDDCR_CPU voltage monitor pin
VDDCR_SOC	Supply voltage for the Northbridge
VDDCR_SOC_SENSE	VDDCR_SOC voltage monitor pin
VDDCR_SOC_S5	Always on SOC power supply
VDDIO_MEM_S3	1.2 V DRAM supply voltage
VDDIO_MEM_S3_SENSE	VDDIO_MEM_S3 voltage monitor pin
VDDP	
VDDP_SENSE	VDDP voltage monitor pin
VDDP_S5	
VDD_18	1.8 V supply voltage
VDD_18_S5	Always on 1.8 V supply voltage
VDD_33	3.3 V supply voltage
VDD_33_S5	Always on 3.3 V supply voltage
VSS	Ground
VSS_SENSE_A	VSS sense pin for voltage regulator
VSS_SENSE_B	VSS sense pin for voltage regulator
CORETYPE[1:0]	Processor Core Type Indicator
AM4R1	Processor Family Revision Identifier

RSVD	Reserved
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